

Basic Principles of Medicine 1
Module: Cardiovascular, Pulmonary,
Renal
Lecture: CPR 52

Title: Acid Base Disorders

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Acknowledgements: Dr. Sharmila Upadhya
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Lecture: Acid base Disorders

SOM.MK.I.BPM1.3.CPR.1.INTG. 0017	Describe processes that lead to respiratory acid-base disturbances (acidosis and alkalosis) and identify common causes for each.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0018	Describe processes that lead to metabolic acid-base disturbances (acidosis and alkalosis). Identify common causes for each including (diabetic ketoacidosis, lactic acidosis, diarrhea, acetazolamide treatment and Vomiting, pyloric stenosis).
SOM.MK.I.BPM1.3.CPR.1.INTG. 0019	Define and calculate the anion gap and distinguish between increased and normal anion gap in metabolic acidosis.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0020	Describe the derivation of the Davenport diagram and identify how it can be used to determine the acid base status in individuals.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0021	Differentiate between metabolic/respiratory acidosis and alkalosis using the Davenport diagram.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0022	Analyze the compensatory mechanisms that are active in the four acid base disorders and how compensation is represented on the Davenport diagram.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0023	Identify simple and mixed metabolic and respiratory acid –base disturbances using the Davenport diagram.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0024	Distinguish between acute and chronic respiratory disturbances. Describe the causes, compensatory responses for: Acute and Chronic Respiratory acidosis – Causes of hypoventilation. Acute and Chronic Respiratory alkalosis – Causes of hyperventilation, high altitude adaptations.
SOM.MK.I.BPM1.3.CPR.1.INTG. 0025	Interpret ABG data (pH, PCO ₂ and HCO ₃ ⁻) and solve problems related to disorders of acid base balance using Davenport diagram. From blood values, identify simple and mixed metabolic and respiratory acid-base disturbances.

Recommended Reading

- Medical Physiology: Principles for Clinical Medicine, R. Rhoades & D. Bell; 6th ed.; Chapter 24, pp. 533-555; [Acid–Base Homeostasis | Medical Physiology: Principles for Clinical Medicine, 6e | Premium Basic Sciences | Health Library \(oclc.org\)](#)
- Ref: Mark's Essentials of Biochemistry: Chapter 4; [Water, Acids, Bases, and Buffers | Marks' Basic Medical Biochemistry: A Clinical Approach, 6e | Premium Basic Sciences | Health Library \(oclc.org\)](#)
- Reference: Vander's Renal Physiology; Chapter 9: Acid base balance; [Regulation of Acid-Base Balance | Vander's Renal Physiology, 10th Edition | AccessMedicine | McGraw Hill Medical \(oclc.org\)](#)
- Additional problems: <http://www.scribd.com/doc/6442924/ABG-Interpretation>
Slide 17 onwards
- http://www.mhhe.com/biosci/ap/ap_casestudies/cases/ap_case14.html
- Practice questions folder for the week
- Small group case

Assessment of acid-base status

- Clinical assessment of acid base status is done with help of Arterial blood gas (ABG) analysis
- Arterial blood gas (ABG) analysis: Arterial pH, HCO_3^- , and PaCO_2
- Normal levels
 - pH: 7.35-7.45
 - PaCO_2 (PCO_2): 33-45 mmHg
 - HCO_3^- : 22-28 mmol/L (mEq/L)

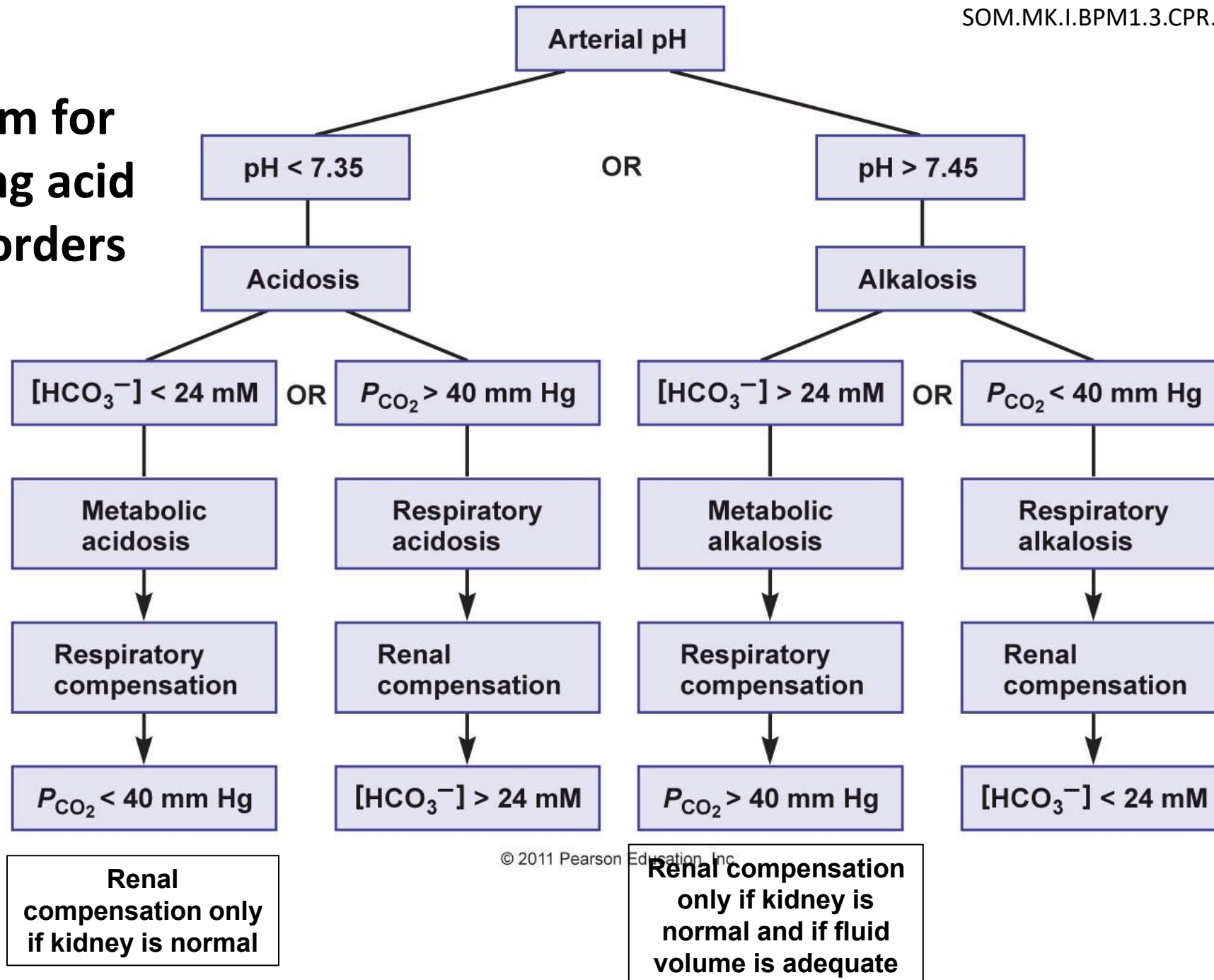


**Make sure you look
at ALL three
parameters AND
NOT at one only!!!!**

Classification of acid-base disorders

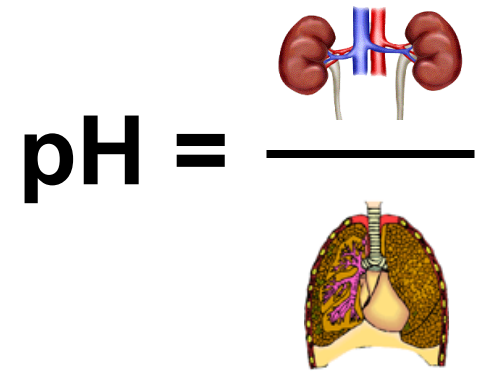
- Acidosis is a condition in which the pH is less than 7.35
- Alkalosis is a condition in which the pH is greater than 7.45
- Respiratory acid base disorders: Dysfunction of the respiratory system and characterized by **primary changes in PaCO_2**
- Metabolic acid base disorders: Metabolic or renal disorders and characterized by **primary changes in $[\text{HCO}_3^-]$**

Algorithm for identifying acid base disorders



Compensatory mechanisms and ratio of $\text{HCO}_3^-/\text{PaCO}_2$

- If the primary disorder is characterized by an **increase in PaCO_2** , compensatory mechanisms will **increase the HCO_3^-** , to bring the ratio towards normal and pH comes closer towards normal
- If the primary disorder is characterized by a **decrease in HCO_3^-** , compensatory mechanisms will **decrease the PaCO_2** , to bring the ratio towards normal and pH comes closer to normal
- **COMPENSATION always in the SAME DIRECTION**



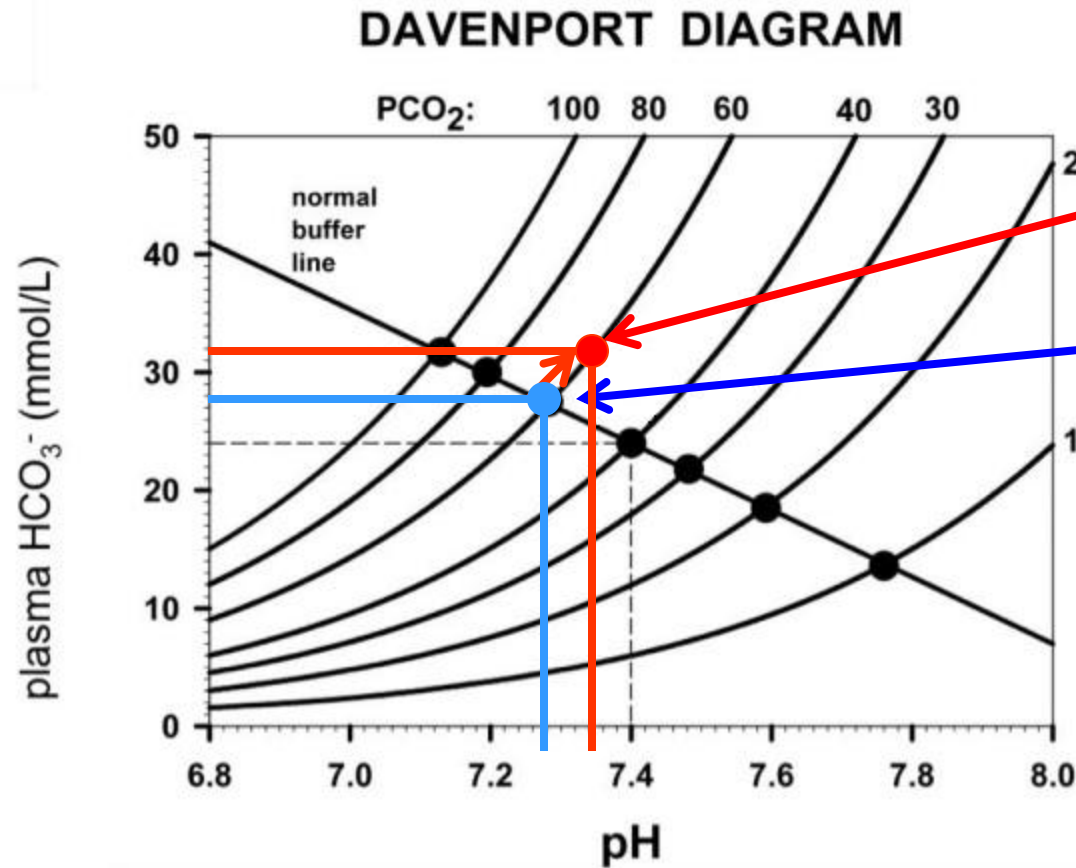
Respiratory acidosis

- A 45-year-old man comes to the physician because of a 1-day history of fever and cough. He is a chronic smoker for the past 20 years. He was diagnosed with chronic obstructive pulmonary disease (COPD) 2 years ago.
- Arterial blood gas analysis shows:
 - pH: 7.23 **Acidosis**
 - PaCO₂: 60 mmHg (primary disturbance) **Respiratory**
 - [HCO₃⁻]: 25 mmol/L **Acute**

Respiratory acidosis

- Characterized by decreased rate of respiration/ decreased lung function/ decreased air entry into the lungs
- As a result, the **CO₂ is NOT washed out (CO₂ accumulation)**, resulting in elevation of PaCO₂ (primary abnormality)
- Remember, CO₂ is an **ACID**, and its accumulation causes respiratory acidosis
- Acute stage,
 - pH Decreased (lower than 7.36)
 - PaCO₂ Elevated (primary disturbance)
 - HCO₃⁻ **is almost normal**

Respiratory acidosis – Davenport diagram



Chronic respiratory acidosis

Acute respiratory acidosis

$$\downarrow pH = \frac{[\text{HCO}_3^-]}{\uparrow PCO_2}$$

Chronic respiratory acidosis

- A 45-year-old man after a 5 days of onset of lung infection (if not treated or treatment not effective)
- Arterial blood gas analysis (Chronic respiratory acidosis)
 - pH: 7.32 (compared to day 1, pH closer to normal) **Acidosis**
 - PCO₂: 60 mmHg (primary change) **Respiratory**
 - [HCO₃⁻]: **31 mmol/L** (compensation by kidneys) **Chronic**

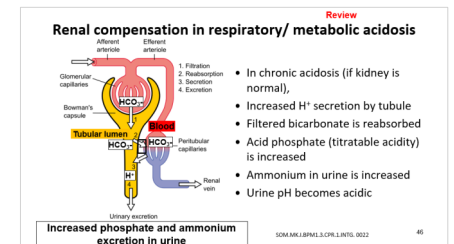
$$\downarrow pH = \frac{\uparrow [HCO_3^-]}{\uparrow PCO_2}$$

Respiratory acidosis

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- Arterial blood gas analysis shows:
 - pH: 7.23 **Acidosis**
 - PaCO₂: 60 mmHg (primary disturbance) **Respiratory**
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Chronic Respiratory acidosis with compensation

- During compensation, renal system comes to the rescue
- **Kidneys** excrete more H^+ , and generate more HCO_3^- , and **serum $[\text{HCO}_3^-]$ levels increase**
- Increased excretion of phosphate (Titratable acid) and ammonium in urine
- Chronic respiratory acidosis (arterial blood gas)
 - pH: Lower than normal (<7.36) - closer to normal pH, when compared to acute stage
 - PaCO_2 : Elevated (Primary defect is still not corrected – respiratory system is still not functioning optimally)
 - $[\text{HCO}_3^-]$: Elevated (due to increased bicarbonate formation by kidney)



Henderson-Hasselbalch equation in respiratory acidosis

$$\text{pH} = \text{pKa} + \log \frac{\uparrow\uparrow [\text{HCO}_3^-]}{0.03 \times \uparrow\uparrow \text{PCO}_2}$$

Compensatory mechanisms by kidney

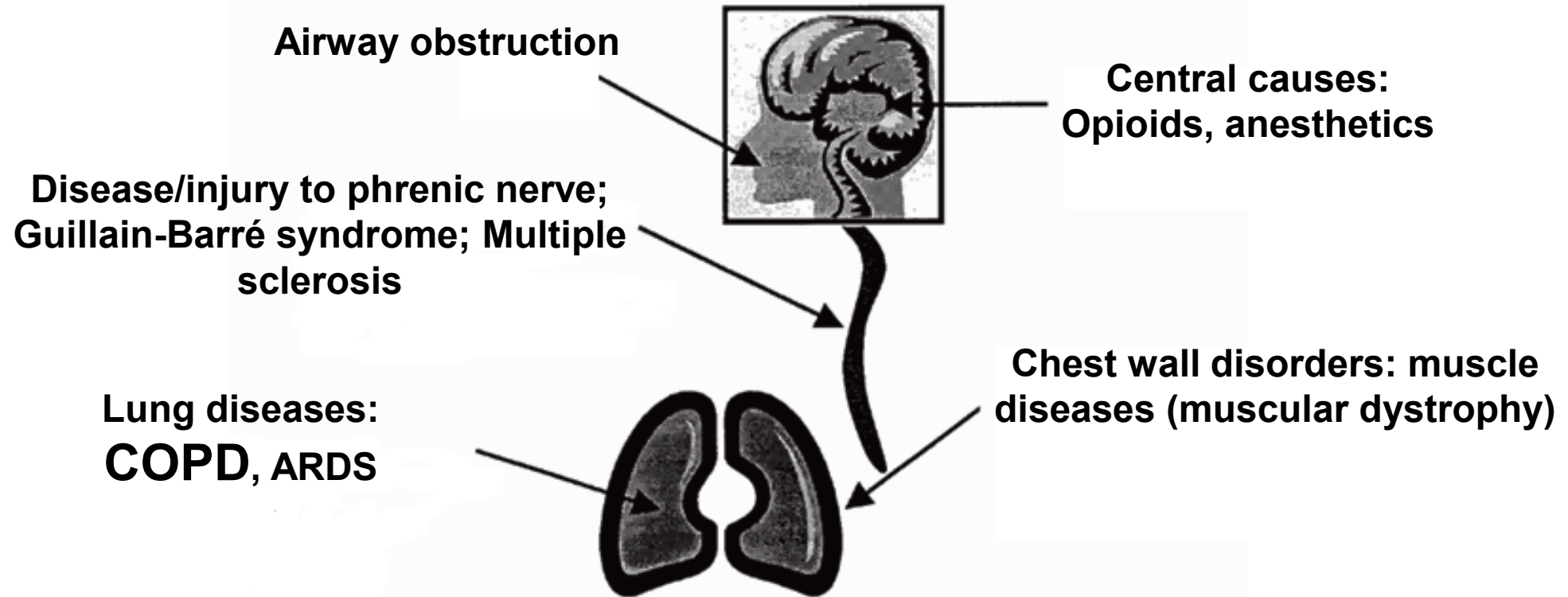
Primary abnormality

- Primary abnormality $\uparrow\uparrow \text{PCO}_2$ due to respiratory disease $\rightarrow$ Fall in pH
- Compensation by renal system, **increases serum $[\text{HCO}_3^-]$** , and ratio of [base]/[acid] returns almost to normal, and pH comes toward normal (it is still below normal)
- **Compensation always in the same direction to bring the ratio towards normal**

Causes of respiratory acidosis

Causes of Hypoventilation and Respiratory Acidosis

Usually associated
with hypoxia



- Inhibit respiratory center (opioids, anesthetics)
- Diseases/ injury of phrenic nerve (supplies diaphragm)
- Lung diseases: **Chronic obstructive pulmonary disease (COPD)**, respiratory distress syndrome in infants or in chronic smokers (Acute Respiratory Distress Syndrome; ARDS)
- Airway obstruction: Laryngeal spasm, Obstructive sleep apnea (obesity)
- Demyelinating disorders: Guillain-Barré syndrome, Multiple sclerosis

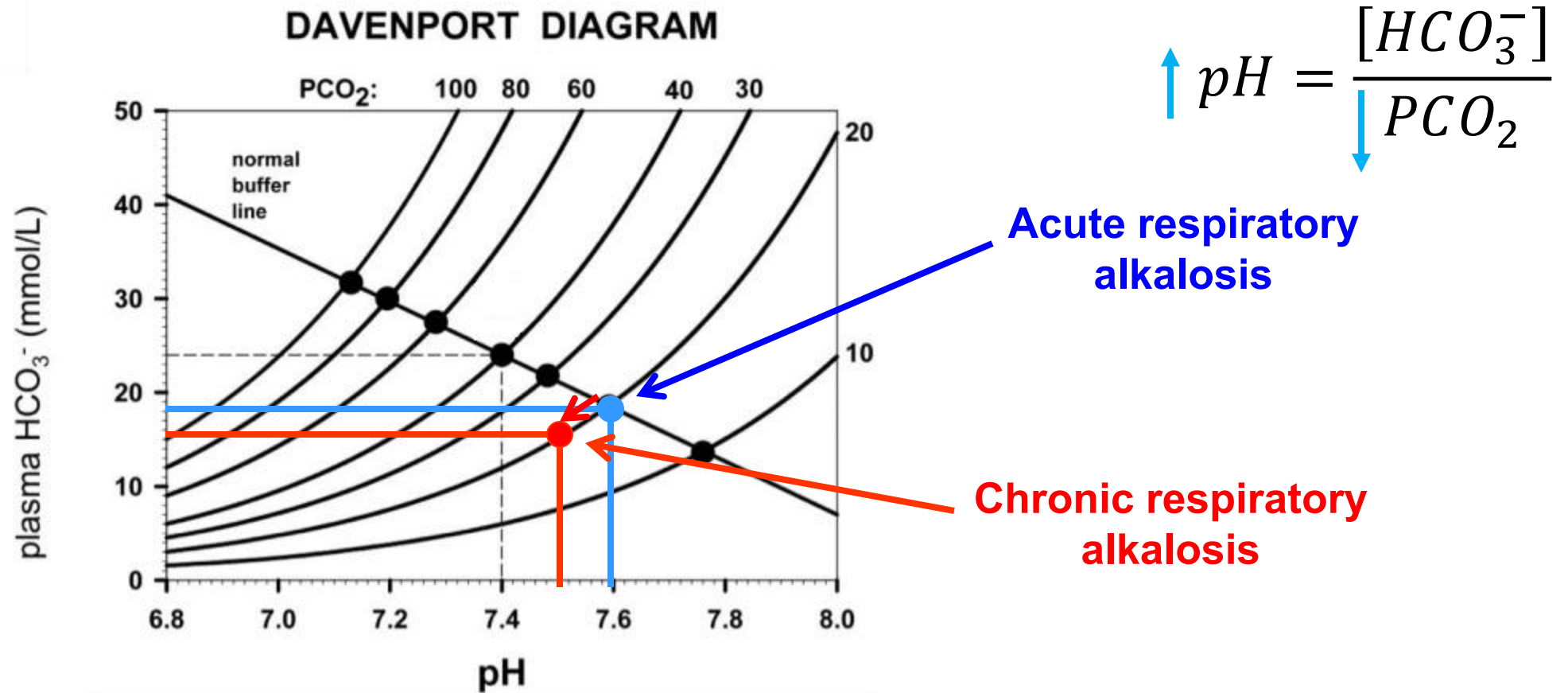
Causes of respiratory acidosis

Cause	Examples	Comments
Inhibition of the medullary respiratory center	•Opiates, barbiturates, anesthetics •Lesions of the central nervous system •Central sleep apnea	
	Oxygen therapy	Inhibition of peripheral chemoreceptors
Disorders of Nerves/ respiratory muscles	Guillain-Barré syndrome, polio, amyotrophic lateral sclerosis (ALS), multiple sclerosis	
Airway obstruction	•Aspiration •Obstructive sleep apnea •Laryngospasm	
Disorders of gas exchange	•Acute respiratory distress syndrome (ARDS) •Chronic obstructive pulmonary disease (COPD) •(Pneumonia) •Pulmonary edema	↓ Exchange of CO ₂ between pulmonary capillary blood and alveolar gas

Acute Respiratory alkalosis

- A 25-year-old student has been out camping in the Himalaya (high altitude). She is hyperventilating.
- Hypoxia (high altitude) stimulates the respiratory center
- The second day of her stay, her arterial blood gas results show
 - pH: 7.56 **Alkalosis**
 - PaCO₂: 25 mmHg **Respiratory**
 - [HCO₃⁻]: 22 mmol/L **Acute**

Respiratory alkalosis – Davenport diagram



Acute Respiratory alkalosis

- Respiratory alkalosis is characterized by an increase in rate of respiration (hyperventilation) → Increased washout of CO_2 → $\downarrow\downarrow \text{PCO}_2$ (primary disturbance) → Higher pH
- Acute respiratory alkalosis, arterial blood gas studies:
 - pH is increased (greater than 7.44)
 - PaCO_2 is $\downarrow\downarrow$ (<35 mmHg)
 - $[\text{HCO}_3^-]$ is almost normal

Chronic respiratory alkalosis

- She is observed 7 days later, and an arterial blood gas analysis is done

- Arterial blood gas analysis after 7 days

- pH: 7.47 (compared to day 2, it is closer to normal) **Alkalosis**
- PaCO₂: 25 mmHg (primary disturbance) **Respiratory**
- [HCO₃⁻]: **18 mmol/L** (renal compensation and increased HCO₃⁻ excretion in urine) **Chronic**

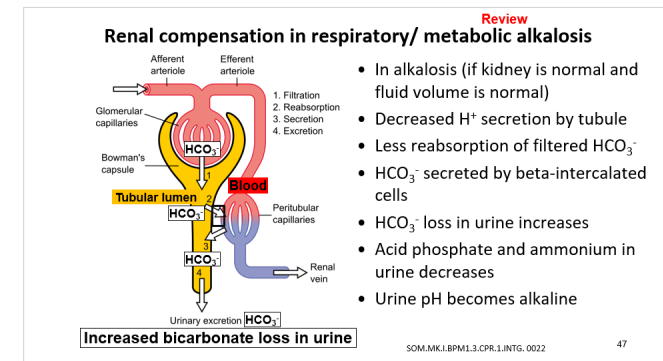
$$\uparrow pH = \frac{\downarrow [HCO_3^-]}{\downarrow PCO_2}$$

Acute Respiratory alkalosis

- A 25-year-old student has been out camping in the Himalaya (high altitude). She is hyperventilating.
- Hypoxia (high altitude) stimulates the respiratory center
- The second day of her stay, her arterial blood gas results show
 - pH: 7.56 **Alkalosis**
 - PaCO₂: 25 mmHg **Respiratory**
 - [HCO₃⁻]: 22 mmol/L **Acute**

Chronic respiratory alkalosis

- The **renal system** tries to bring the pH back towards normal
- Kidneys do NOT secrete H^+ into urine (inhibition of carbonic anhydrase)
- **Increased urinary HCO_3^- excretion (Beta intercalated cells)** and pH of urine becomes alkaline, and **serum HCO_3^- falls**
- Chronic respiratory alkalosis,
 - pH: Higher than normal - closer to normal pH, when compared to acute stage
 - PCO_2 : Decreased (primary defect is still not corrected; **Hypoxia** → Hyperventilation)
 - **$[HCO_3^-]$: Decreased** (due to renal bicarbonate loss)



Henderson-Hasselbalch equation in respiratory alkalosis

$$\text{pH} = \text{pK}_a + \log \frac{\downarrow\downarrow [\text{HCO}_3^-]}{0.03 \times \downarrow\downarrow \text{PaCO}_2}$$

Compensatory mechanisms by kidney

Primary abnormality

- Primary abnormality is $\downarrow\downarrow \text{PaCO}_2$ due to hyperventilation, which results in rise in pH (alkalosis)
- Compensation by renal system, **decreases serum $[\text{HCO}_3^-]$** , and ratio of [base]/[acid] returns almost to normal, and pH comes toward normal (it is still above normal)
- **Compensation always in the same direction to bring the ratio towards normal**

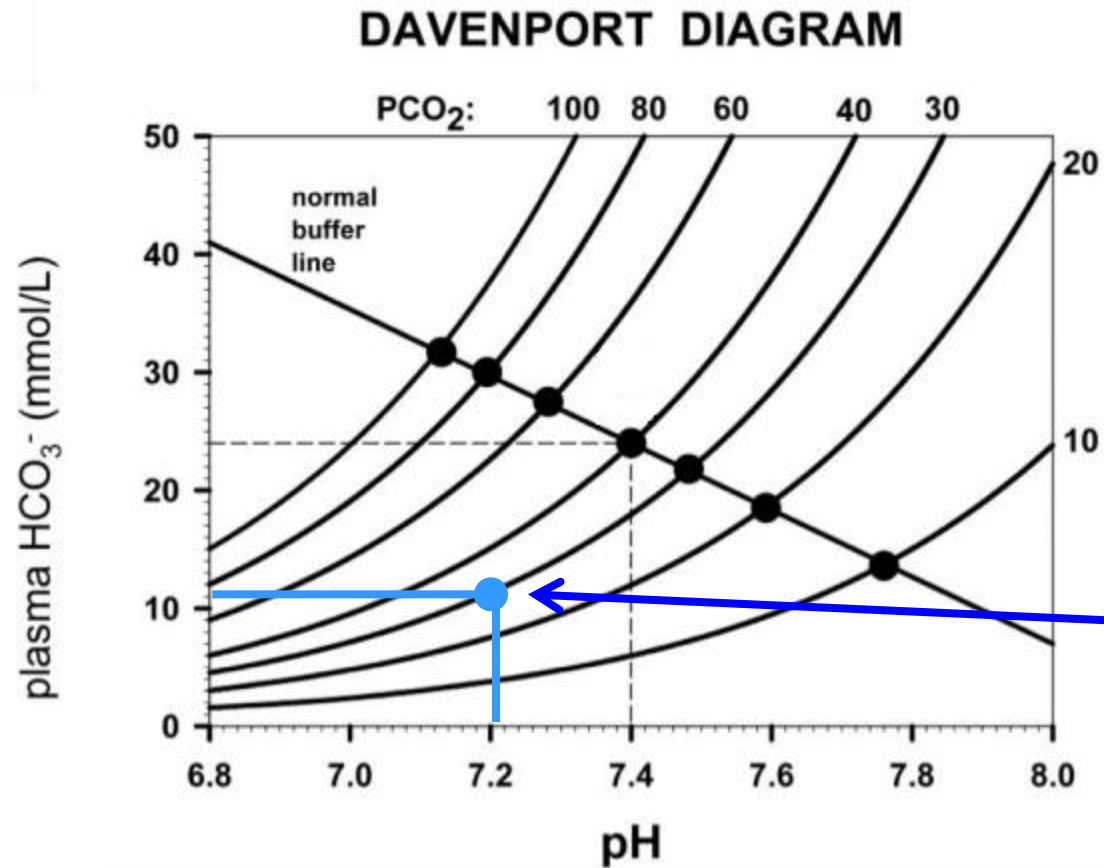
Causes of respiratory alkalosis

- Causes of **hyperventilation**
 - **Anxiety, fever, Somatic symptom disorder**
 - **High altitude: Hypoxia** stimulates the respiratory center and increases rate of respiration. When a person stays for a long time at elevated altitude, renal compensatory mechanisms are active and $[\text{HCO}_3^-]$ levels fall
 - Refer to hemoglobin lecture: Identify changes in oxygen dissociation curve when a person is adapted to high altitude
 - **Mechanical ventilation (Overventilation)**
 - **Drugs that stimulate respiratory center (salicylate)**

Metabolic acidosis

- A 26-year-old woman is brought to the ER in a state of altered consciousness. She is a known type-I diabetic on insulin.
- Arterial blood gas analysis on admission
 - pH: 7.21 **Acidosis**
 - PaCO₂: 30 mmHg **Compensation**
 - [HCO₃⁻]: 12 mmol/L **Metabolic**
 - HCO₃⁻ lost due to buffering ketone bodies – primary abnormality
 - Na⁺: 140 mEq/L
 - Cl⁻: 100 mEq/L
 - K⁺: 5.3 mEq/L
- Plasma glucose: 500mg/dL
- Dipstick test with urine for ketone bodies and glucose is positive
- She is hyperventilating (Kussmaul breathing)

Metabolic acidosis – Davenport diagram



$$\downarrow pH = \frac{\downarrow [\text{HCO}_3^-]}{\downarrow \text{PCO}_2}$$

Metabolic acidosis

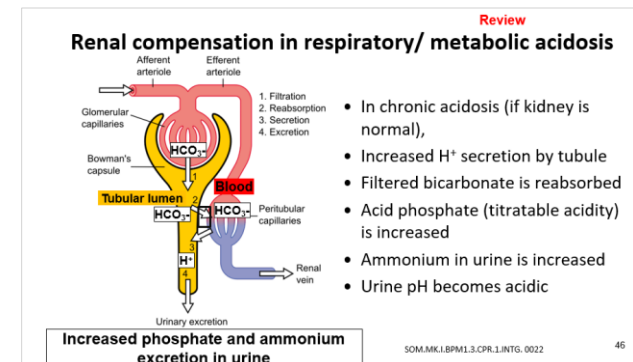
Note: No acute/ chronic for metabolic disorders

Metabolic acidosis

- Characterized by $\downarrow\downarrow[\text{HCO}_3^-] \rightarrow \text{Low pH}$
- $[\text{HCO}_3^-]$ is low due to
 - **Increased nonvolatile acids** (HCO_3^- lost by buffering)
 - **OR Increased losses of HCO_3^-**
- Acidosis (Low pH) stimulates respiratory center $\rightarrow$ **Hyperventilation**
 $\rightarrow$ Increased washout of $\text{CO}_2 \rightarrow$ **Fall in PaCO_2 (Remember, respiratory center responds within minutes; compensation is immediate)**
 - **Compensatory hyperventilation $\rightarrow$ Low PaCO_2**

Metabolic acidosis – Role of kidney in compensation

- If kidney is functioning, it can also compensate to increase H^+ excretion, increase the formation of new HCO_3^- .
- Renal system takes hours to days to respond.
- Carbonic anhydrase active $\rightarrow$ Increased H^+ secretion in renal tubules
 - Increased excretion of acid phosphate (titratable acid) and **ammonium** in urine
- However, until the cause of acidosis is treated, pH does not come back to normal



Henderson-Hasselbalch equation in metabolic acidosis

$$\text{pH} = \text{pKa} + \log \frac{\downarrow\downarrow [\text{HCO}_3^-]}{0.03 \times \downarrow\downarrow \text{PCO}_2}$$

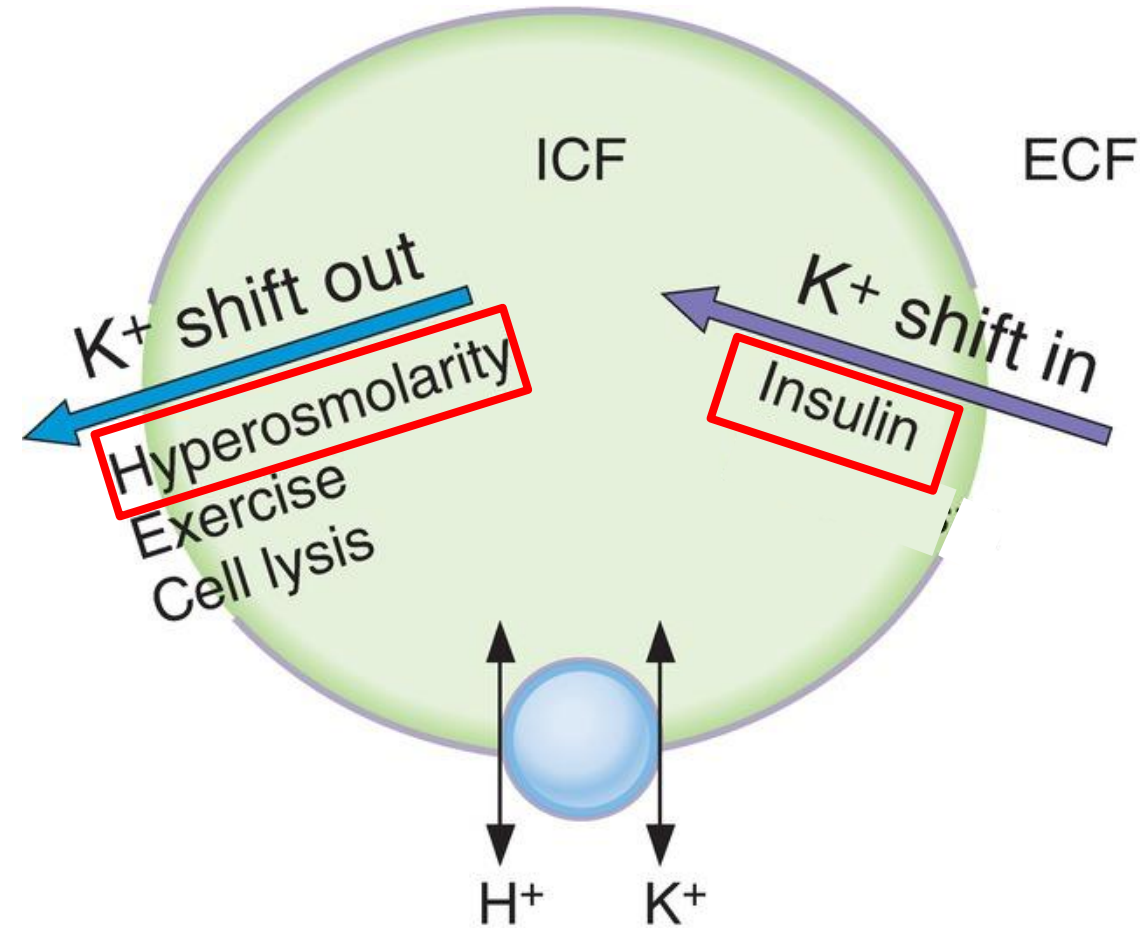
Primary abnormality

Compensatory mechanisms by lungs

- Primary abnormality is $\downarrow\downarrow \text{HCO}_3^-$, which results in a fall in pH
- Compensation by respiratory system, results in hyperventilation, and **decreases PaCO₂**, and ratio of [base]/[acid] returns almost to normal, and pH comes toward normal (it is still below normal)
- **The kidneys can compensate too, if they are normal – they try to increase the serum HCO₃⁻**

H⁺ and K⁺ interrelations

- K⁺ (major intracellular cation)
- K⁺ and H⁺ are closely related
- **Diabetic ketoacidosis:**
 - Plasma glucose: 500mg/dL; Serum K⁺: 5.3 mEq/L
 - **Insulin deficiency and hyperglycemia (hyperosmolarity)**
 - → K⁺ shift from ICF to ECF
 - → Increased serum K⁺ (Hyperkalemia)



Serum K⁺ and diabetic ketoacidosis

- Serum K⁺ increased (K⁺: 5.3mEq/L)
- Plasma glucose: 500mg/dL (ECF **Hyperosmolarity** causes K⁺ to shift from ICF to ECF with osmotic fluid shift)
- **Insulin deficiency** in type-1 diabetes mellitus prevents K⁺ entry into ICF (K⁺ levels in ECF increase) – Remember, Insulin activates the Na-K ATPase
- What happens to serum K⁺ when you treat the patient with insulin?

Metabolic acidosis

- A 26-year-old woman is brought to the ER in a state of altered consciousness. She is a known type-1 diabetic on insulin.
- Arterial blood gas analysis on admission
 - pH: 7.21 **Acidosis**
 - PaCO₂: 30 mmHg **Compensation**
 - [HCO₃⁻]: 12 mmol/L **Metabolic**
 - HCO₃⁻ lost due to buffering ketone bodies – primary abnormality
 - Na⁺: 140 mEq/L
 - Cl⁻: 100 mEq/L
 - K⁺: 5.3 mEq/L
- Plasma glucose: 500mg/dL
- Dipstick test with urine for ketone bodies and glucose is positive
- She is hyperventilating (Kussmaul breathing)

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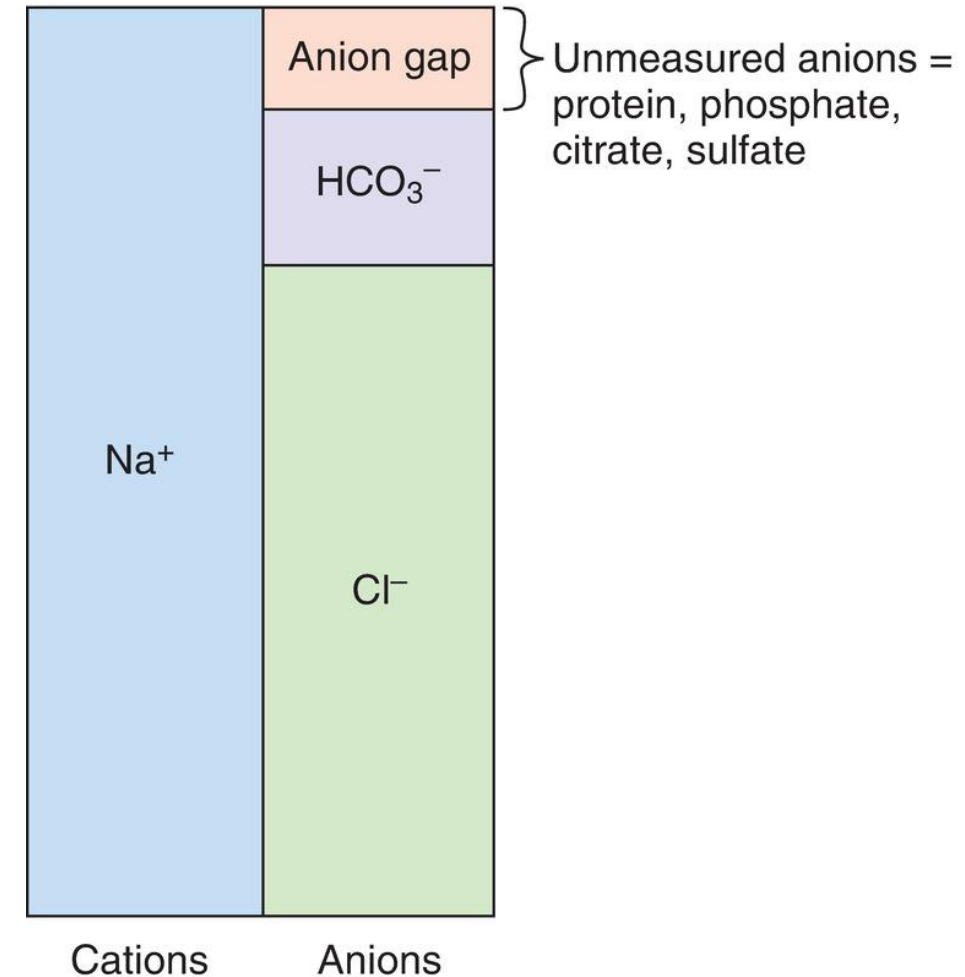
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 - → K⁺ shift from ICF to ECF
 - → Increased serum K⁺ (Hyperkalemia)

SOM.MK.I.BPM1.3.CPR.1.INTG.0015

Anion Gap

- Anion Gap = Cations – Sum of Anions
- Anion gap = $[\text{Na}^+] - \{[\text{Cl}^-] + [\text{HCO}_3^-]\}$
- Normal Anion Gap is 8-14 mEq/L
- Due to presence of **unmeasured anions**
- Unmeasured anions are **phosphate, sulfate, lactate, ketone bodies** and protein
- In some forms of **metabolic acidosis**, there is increase in unmeasured anions, resulting in increased anion gap



[Anion gap explained-https://www.youtube.com/watch?v=sQnEFVNrY74](https://www.youtube.com/watch?v=sQnEFVNrY74)

(Note: K⁺ is NOT included in anion gap calculation)

Calculate anion gap

- Arterial blood gas analysis on admission
 - pH: 7.21
 - PaCO₂: 30 mmHg
 - [HCO₃⁻]: 12 mmol/L
 - Na⁺: 140 mEq/L
 - Cl⁻: 100 mEq/L
 - K⁺: 5.3 mEq/L
- $140 - 112 = 28 \text{ mEq/L}$ (Normal: 8-14 mEq/L)
- **High anion gap** metabolic acidosis
- In ketoacidosis, ketone bodies are the **unmeasured anions**

Metabolic acidosis (normal anion gap)

- A 3-year-old child with **severe diarrhea**, has the following reports:
- Arterial blood gas analysis on admission
 - pH: 7.31
 - PaCO₂: 32 mmHg
 - [HCO₃⁻]: 16 mmol/L (bicarbonate lost in intestinal secretions – primary abnormality)
 - Na⁺: 138 mEq/L
 - Cl⁻: 109 mEq/L
- Calculate anion gap: $138 - 125 = 13$ mEq/L (Normal: 8-14 mEq/L)
- **Normal anion gap (hyperchloremic) metabolic acidosis**

Causes of metabolic acidosis

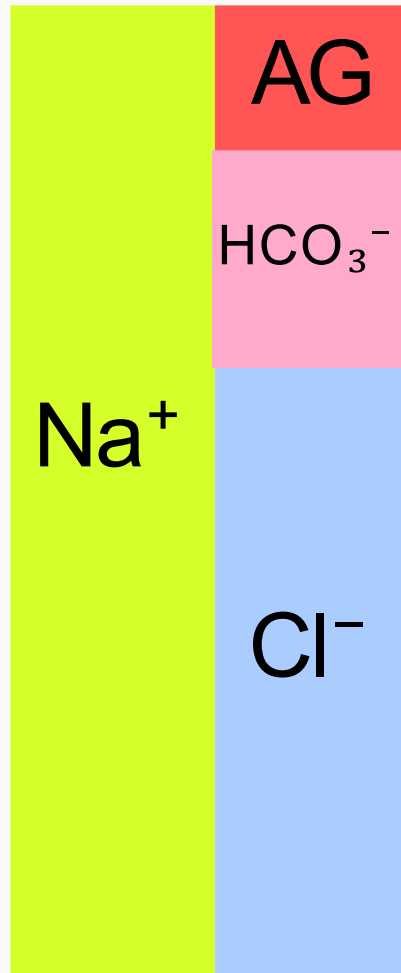
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graph TD; A[Causes of metabolic acidosis] --> B[Increased production of non-volatile acids (High anion gap metabolic acidosis) – Increased unmeasured anions]; A --> C[Increased loss of HCO3- (base) – Normal anion gap metabolic acidosis – Hyperchloremic acidosis]; B --> D[Diabetic ketoacidosis (increased ketone body production), Lactic acidosis (increased lactate production), Chronic renal failure (decreased excretion of sulfate, phosphate), Methanol and ethylene glycol poisoning]; C --> E[Diarrhea (increased loss of HCO3- rich intestinal secretions), Renal tubular acidosis (failure to secrete H+ and reabsorb HCO3-), Acetazolamide treatment (inhibits renal carbonic anhydrase)];
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**Increased production of non-volatile acids
(High anion gap metabolic acidosis) –
Increased unmeasured anions**

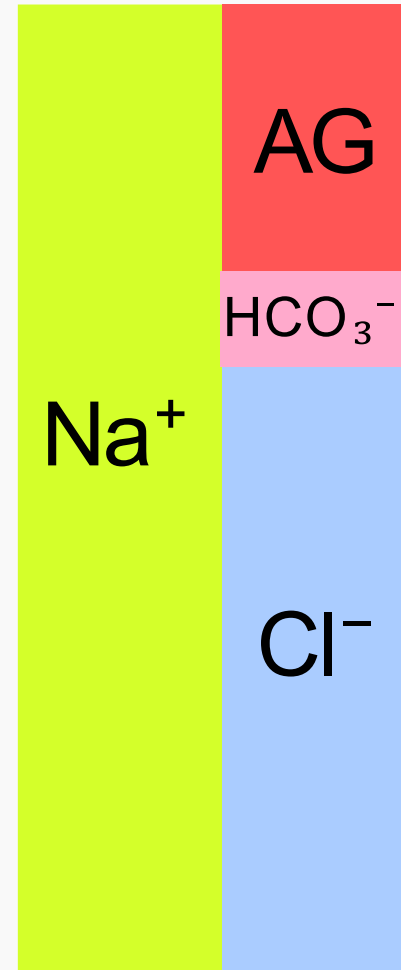
- Diabetic ketoacidosis (increased ketone body production)
- Lactic acidosis (increased lactate production)
- Chronic renal failure (decreased excretion of sulfate, phosphate)
- Methanol and ethylene glycol poisoning

**Increased loss of HCO_3^- (base) –
Normal anion gap metabolic acidosis
– Hyperchloremic acidosis**

- Diarrhea (increased loss of HCO_3^- rich intestinal secretions)
- Renal tubular acidosis (failure to secrete H^+ and reabsorb HCO_3^-)
- Acetazolamide treatment (inhibits renal carbonic anhydrase)

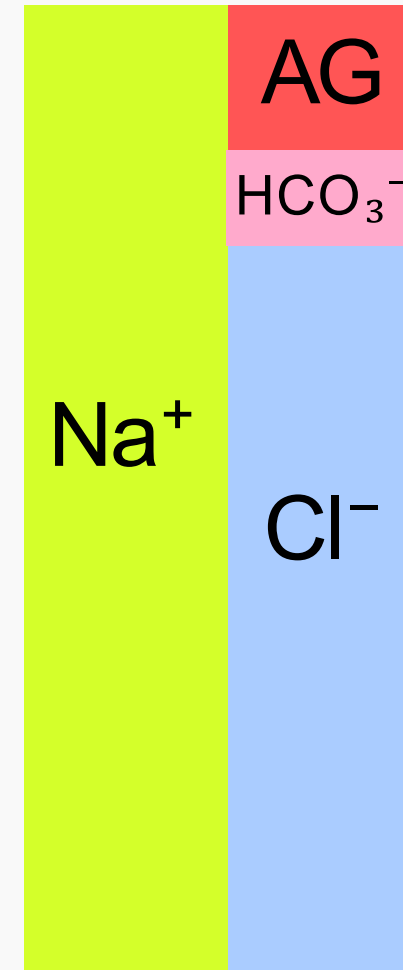


Anion Gap



HAGMA

Diabetic ketoacidosis;
Lactic acidosis
Poisons

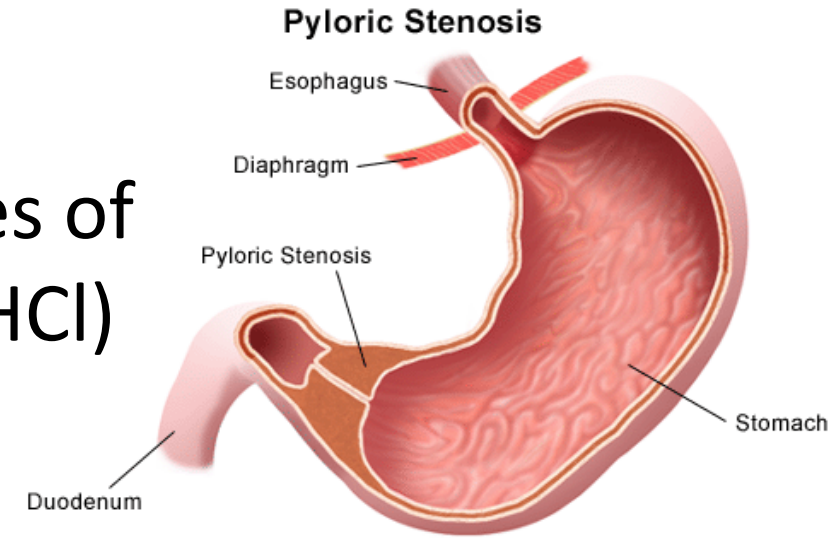


NAGMA

Diarrhea;
Renal tubular acidosis

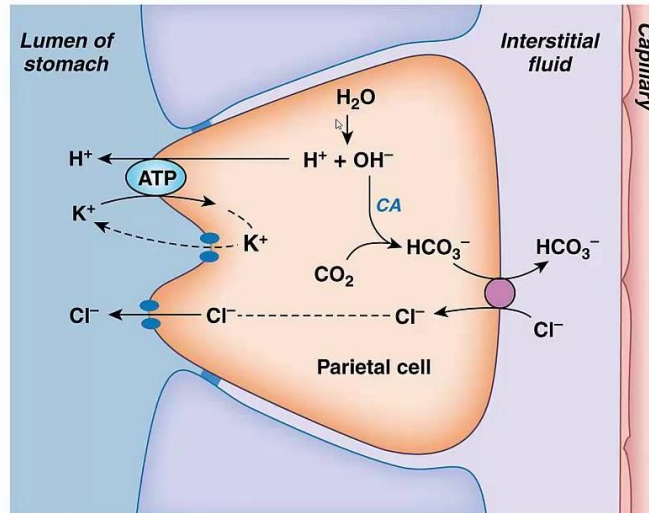
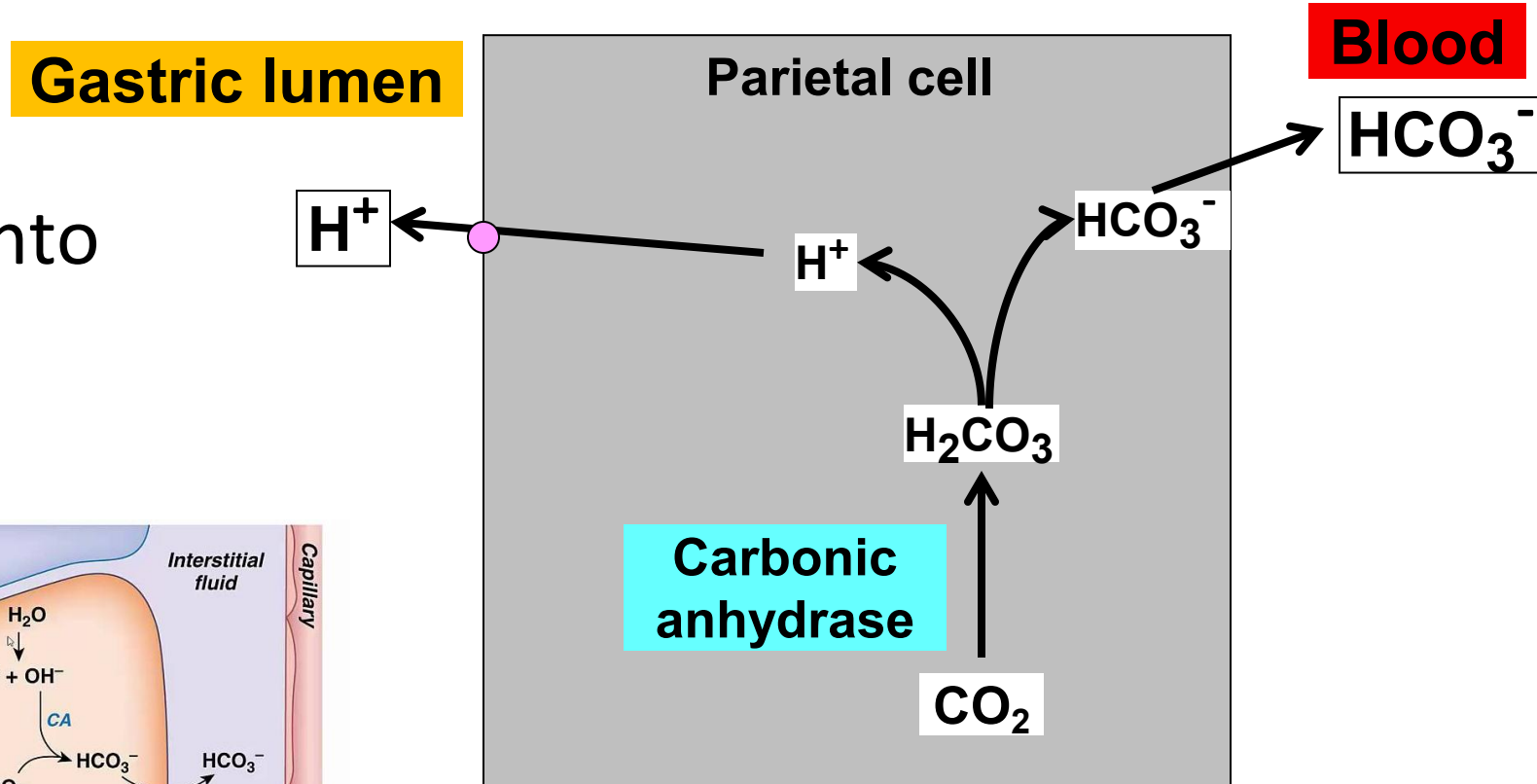
Case on metabolic alkalosis

- A 6-week-old boy has been brought for consultation. He has had repeated episodes of **projectile** vomiting (loss of stomach acid: HCl) for the past two days.
- Arterial blood gas analysis on admission
 - pH: 7.51 **Alkalosis**
 - PaCO₂: 48 mmHg **Compensation**
 - [HCO₃⁻]: 38 mmol/L (primary abnormality) **Metabolic**



Loss of H^+ in vomiting/nasogastric suction

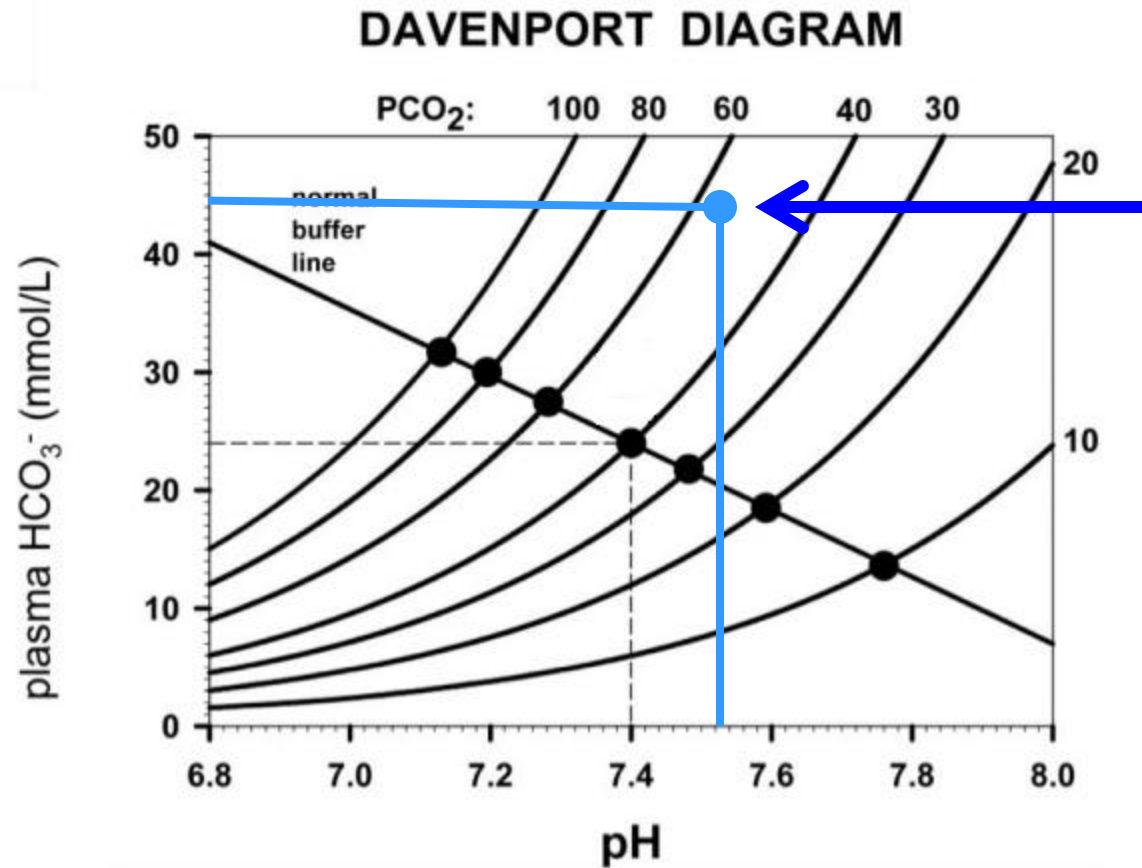
- Parietal cells contain Carbonic anhydrase
- For every H^+ pumped into the gastric lumen, one HCO_3^- gained by blood



Metabolic alkalosis

- Characterized by **increased $[\text{HCO}_3^-]$** (relative excess of HCO_3^-)
- Increase in the pH (alkalosis) **inhibits** the respiratory center, → **Decrease** in rate of respiration (**hypoventilation**) → Decreased washout of CO_2 (CO_2 retention) → **$\uparrow \text{PaCO}_2$ (Compensatory hypoventilation)**
- Metabolic alkalosis
 - pH is higher than normal
 - $[\text{HCO}_3^-]$ is increased (primary abnormality)
 - PaCO_2 is increased (compensatory hypoventilation)

Metabolic alkalosis – Davenport diagram



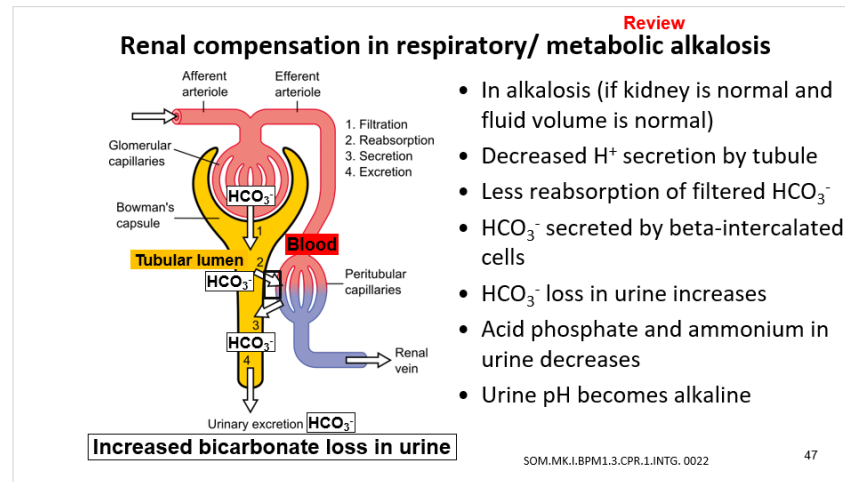
Metabolic alkalosis

$$\uparrow pH = \frac{\uparrow [\text{HCO}_3^-]}{\uparrow PCO_2}$$

Note: No acute/ chronic for metabolic disorders

Metabolic alkalosis – Renal compensation

- Renal system compensates (if the blood volume is normal), by excreting more HCO_3^- (alkaline urine)
- Low tubular H^+ secretion
- Beta-intercalated cells secrete bicarbonate into urine



Henderson-Hasselbalch equation in metabolic alkalosis

$$\text{pH} = \text{pK}_a + \log \frac{\uparrow\uparrow [\text{HCO}_3^-]}{0.03 \times \uparrow\uparrow \text{PaCO}_2}$$

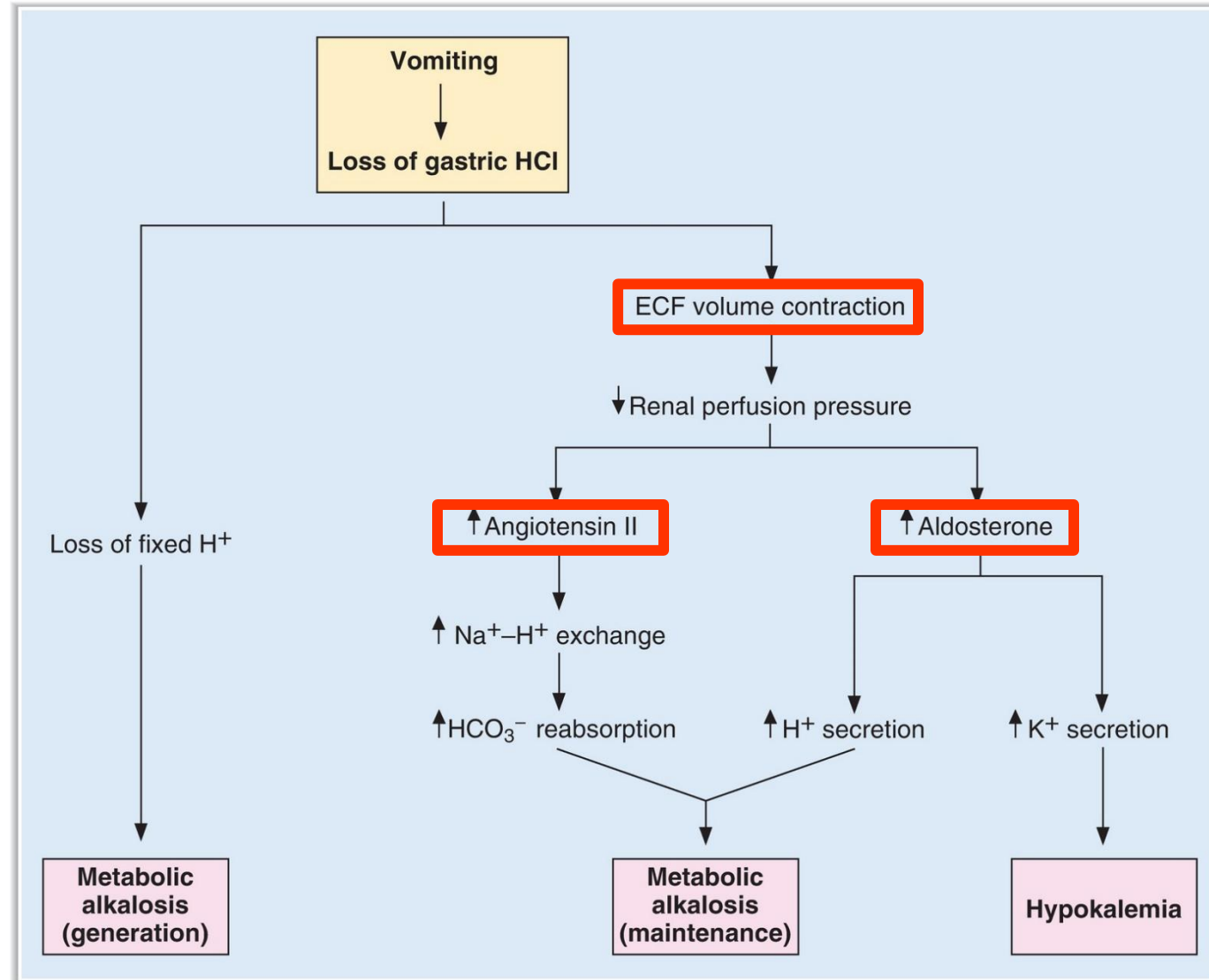
Primary
abnormality

Compensatory
mechanisms
by lungs

- Primary abnormality is $\uparrow\uparrow \text{HCO}_3^-$, which results in a rise in pH
- Compensation by respiratory system, **increases PaCO₂**, and the ratio of [base]/[acid] returns almost to normal, and pH comes toward normal (it is still above normal)
- **Kidneys can compensate if fluid volume is adequate**

Renal response in metabolic alkalosis **with volume contraction**

- **Volume depletion (low blood pressure)**, renal compensation for alkalosis may not be seen!!!
- **RAAS active**
- Kidneys conserve sodium and secrete protons in urine (Na-H^+ exchanger active) → **'Acidic urine'** – **Paradoxic aciduria**
- **TIP:** Restore blood volume before treating the metabolic alkalosis



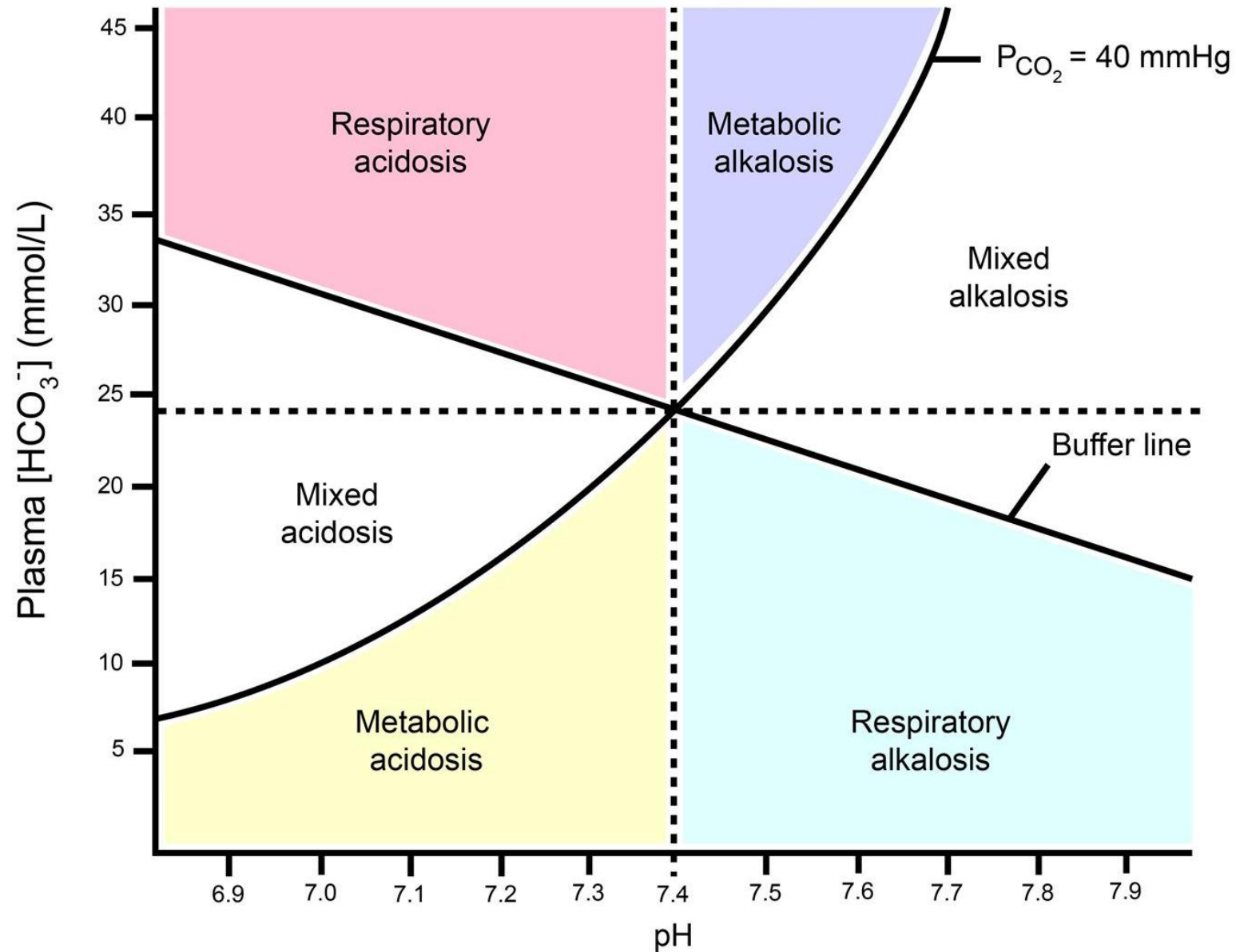
Causes of metabolic alkalosis

- Vomiting, pyloric stenosis (Gastric outlet obstruction)
 - Loss of acidic contents of stomach, results in relative HCO_3^- excess
- Nasogastric suction: A tube removes acidic contents of stomach
- Excessive consumption of antacids
- Primary hyperaldosteronism causes hypokalemia and metabolic alkalosis (explained in previous lecture)
- Hypokalemia (explained in previous lecture)
- Diuretics (cause bulk flow of Na^+ in tubules and ECF volume contraction) → Active RAAS → K^+ secretion → Hypokalemia → H^+ secretion → Alkalosis (Thiazide and loop diuretics)

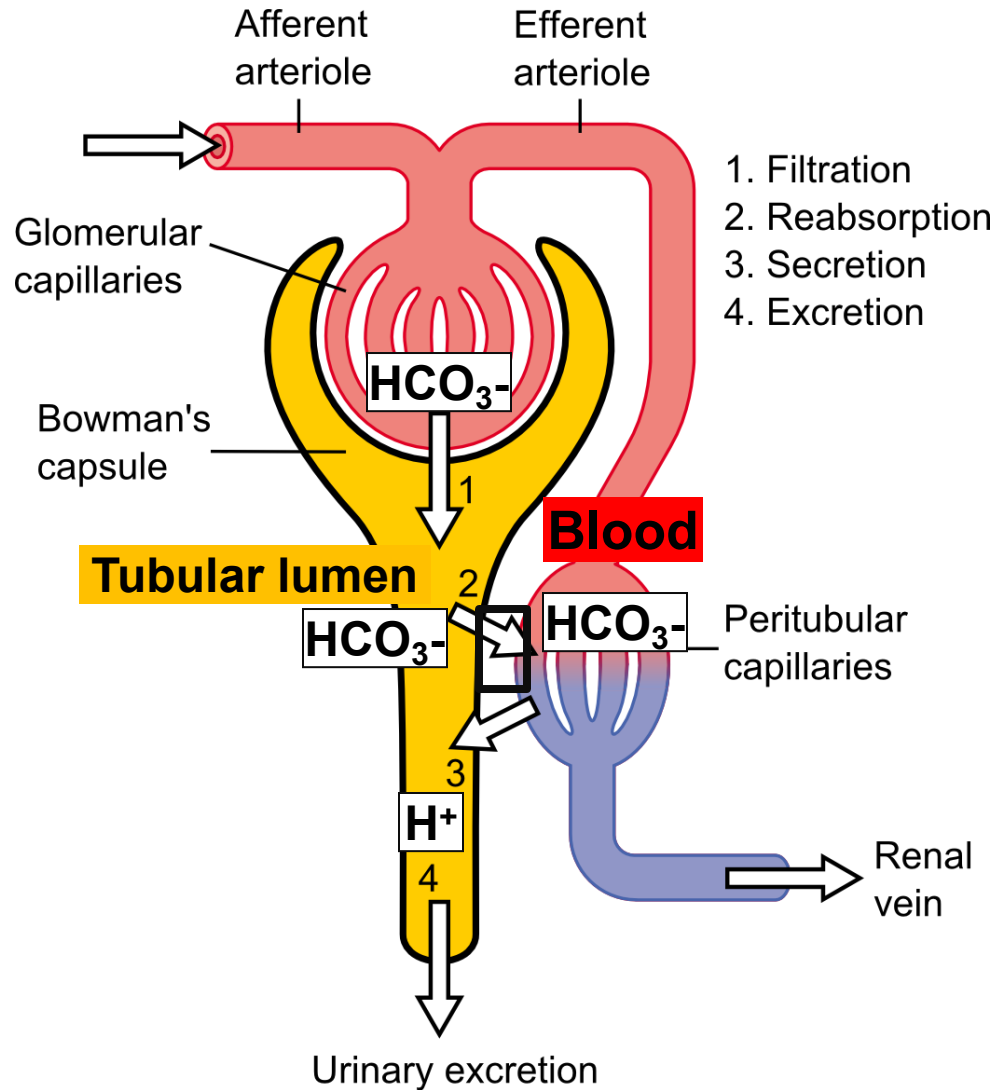
Summary

- First does the patient have an acidosis or alkalosis
 - Look at pH
- Next, what is the primary problem? Is it metabolic or respiratory?
 - Look at PCO_2 and pH, (**ROMS = Respiratory Opposite; Metabolic Same**)
 - If the PCO_2 change is the opposite direction of the pH change, then it is a respiratory disorder
 - Eg: If pH- low and PCO_2 - high (respiratory)
 - If the PCO_2 change is in the same direction as pH change, then it is a metabolic disorder
 - Eg: If pH-low and PCO_2 : low (metabolic); confirm by looking at HCO_3
- Check whether chronic or not (for respiratory acid base disorders)
 - Look at other parameter (HCO_3^-)
 - If there is a change in the same direction (implies chronic state)

Davenport Diagram



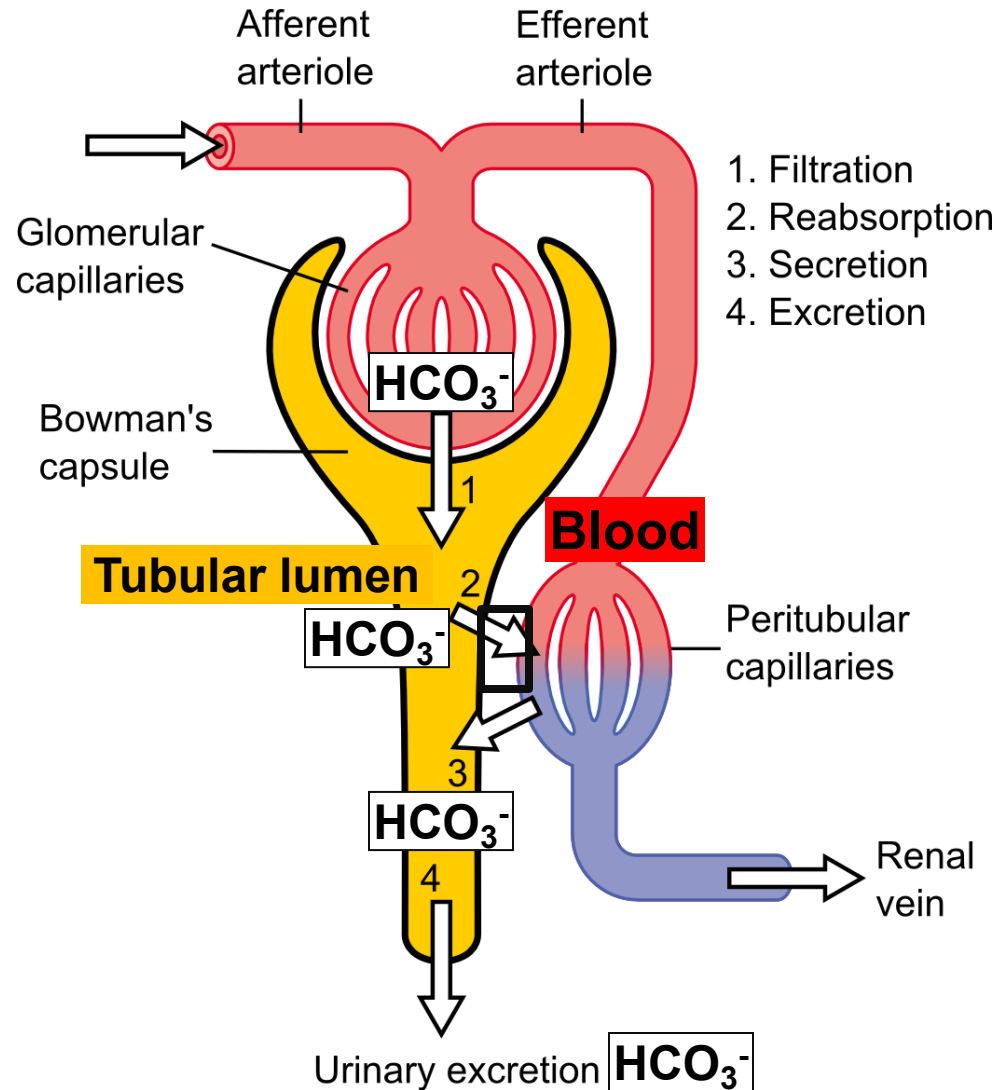
Renal compensation in respiratory/ metabolic acidosis



- In chronic acidosis (if kidney is normal),
- Increased H^+ secretion by tubule
- Filtered bicarbonate is reabsorbed
- Acid phosphate (titratable acidity) is increased
- Ammonium in urine is increased
- Urine pH becomes acidic

Increased phosphate and ammonium excretion in urine

Renal compensation in respiratory/ metabolic alkalosis



- In alkalosis (if kidney is normal and fluid volume is normal)
- Decreased H^+ secretion by tubule
- Less reabsorption of filtered HCO_3^-
- HCO_3^- secreted by beta-intercalated cells
- HCO_3^- loss in urine increases
- Acid phosphate and ammonium in urine decreases
- Urine pH becomes alkaline

Increased bicarbonate loss in urine

Summary

	Arterial Plasma				
Disturbance	pH = 7.4	[H ⁺] = 40nmol/L	[HCO ₃ ⁻] = 24mEq/L	Paco ₂ = 40mmHg	Compensatory Response
Acute Respiratory acidosis					
Chronic Respiratory acidosis					
Acute Respiratory alkalosis					
Chronic Respiratory alkalosis					
Metabolic acidosis					
Metabolic alkalosis					

Summary: Answers

Directional Changes in Arterial Blood Plasma Values in the **Four Simple Acid–Base Disturbances**
Double arrows indicate the main effect. Brackets denote concentration; $Paco_2$, partial pressure of carbon dioxide.

	Arterial Plasma				
Disturbance	pH = 7.4	[H ⁺] = 40nmol/L	[HCO ₃ ⁻] = 24mEq/L	Paco ₂ = 40mmHg	Compensatory Response
Acute Respiratory acidosis	↓	↑	Normal	↑↑	
Chronic Respiratory acidosis	↓	↑	↑	↑↑	Kidneys increase H ⁺ excretion
Acute Respiratory alkalosis	↑	↓	Normal	↓↓	
Chronic Respiratory alkalosis	↑	↓	↓	↓↓	Kidneys increase HCO ₃ ⁻ excretion
Metabolic acidosis	↓	↑	↓↓	↓	Alveolar hyperventilation; Kidneys increase H ⁺ excretion (if kidneys normal)
Metabolic alkalosis	↑	↓	↑↑	↑	Alveolar hypoventilation; Kidneys increase HCO ₃ ⁻ excretion (if blood volume normal)